



RAGzilla

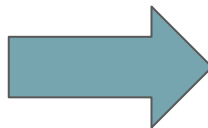
Professional Mini-RAG System for
Question Answering

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Problem Statement

Users struggle to find answers in private PDFs and reports. Keyword search lacks context and precision.



Goal

Build an intelligent PDF QA system using RAG.

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Project Objectives

R

RETRIEVAL

A

AUGMENTATION

G

GENERATION

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Project Overview



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Technologies Used

Library	Purpose
<code>streamlit</code>	Builds the chatbot web interface.
<code>fitz (PyMuPDF)</code>	Extracts text from uploaded PDFs.
<code>faiss</code>	Stores and retrieves document embeddings efficiently.
<code>numpy</code>	Handles embedding vectors and data conversion.
<code>re</code>	Performs text cleaning using regular expressions.
<code>SentenceTransformer</code>	Converts text into embeddings.
<code>CrossEncoder</code>	Re-ranks retrieved chunks by relevance.
<code>ollama</code>	Connects to the local LLM for query reformulation and answer generation.

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Result



Mini-RAG PDF Chatbot

Stop Deploy

Upload a PDF and ask questions about it.

Features:

- PDF Question Answering
- FAISS Vector Search
- Multi-Query Retrieval
- Cross-Encoder Re-ranking
- Hallucination Reduction
- "I Don't Know" Detection

Upload a PDF



Drag and drop file here
Limit 200MB per file • PDF

Browse files



Articles.pdf 190.6KB



Ask a question:

what is a preamble

PDF processed successfully. Created 5 chunks.

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Result

PDF processed successfully. Created 5 chunks.

 Get Answer

 Reformulated Queries:

```
▼ [
  0 : "what is a preamble
      "
  1 : "What is the introductory statement in a formal document?"
  2 : "Can you explain the purpose of an opening paragraph in a written text?"
  3 : "What is the first section or introduction to a speech, document, or legal agreement?"
]
```



Answer

A preamble is an introductory section that is typically found at the beginning of a document, such as a constitution, treaty, or declaration. It serves as an introduction to the main content of the document, providing background information, explaining the purpose and scope of the document, and setting out the key principles and objectives.



Retrieved Sources

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Result

Answer

[Deploy](#)

A preamble is an introductory section that is typically found at the beginning of a document, such as a constitution, treaty, or declaration. It serves as an introduction to the main content of the document, providing background information, explaining the purpose and scope of the document, and setting out the key principles and objectives.

Retrieved Sources

Source Chunk 1

Universal Declaration of Human Rights Preamble Whereas recognition of the inherent dignity and of the equal and inalienable rights of all members of the human family is the foundation of freedom, justice and peace in the world, Whereas disregard and contempt for human rights have resulted in barbarous acts which have outraged the conscience of mankind, and the advent of a world in which human beings shall enjoy freedom of speech and belief and freedom from fear and want has been proclaimed as the highest aspiration of the common people, Whereas it is essential, if man is not to be compelled to have recourse, as a last resort, to rebellion against tyranny and oppression, that human rights should be protected by the rule of law, Whereas it is essential to promote the development of friendly relations between nations, Whereas the peoples of the United Nations have in the Charter reaffirmed their faith in fundamental human rights, in the dignity and worth of the human person and in the equal rights of men and women and have determined to promote social progress and better standards of life in larger freedom, Whereas Member States have pledged themselves to achieve, in cooperation with the United Nations, the promotion of universal respect for and observance of human rights and fundamental freedoms, Whereas a common understanding of these rights and freedoms is of the greatest importance for the full realization of this pledge, Now, therefore, The General Assembly, Proclaims this Universal Declaration of Human Rights as a common standard of achievement for all peoples and all nations, to the end that every individual and every organ of society, keeping this Declaration constantly in mind, shall strive by teaching and education to promote respect for these rights and freedoms and by progressive measures, national and international, to secure their universal and effective recognition and observance, both among the peoples of Member States themselves and among the peoples of territories under their jurisdiction. Article 1 All human beings are born free and equal in dignity and rights. They are endowed with reason and conscience and should act towards one another in a spirit of brotherhood. Article 2 Everyone is entitled to all the rights and freedoms set forth in this Declaration, without distinction of any kind, such as race, colour, sex, language, religion, political or other opinion, national or social origin, property, birth or other status. Furthermore, no distinction shall be made on the basis of the political, jurisdictional or international status of the country or territory to which a person belongs, whether it be independent, trust, non-self-governing or under any other limitation of sovereignty. Article 3 Everyone has the right to life, liberty and the security of person. Article 4 No one shall be held in slavery or servitude; slavery and the slave trade shall be prohibited in all their forms. Article 5 No one shall be subjected to torture or to cruel, inhuman or degrading treatment or punishment. Article 6 Everyone has the right to

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Thank you